I think you know my background. Please prepare a plan for upskilling me to become job-ready for MAANG companies now and in the future (at least for the next 5 years), considering the demand for AI skills required by companies in India, the US, and Australia.

Based on your background, you are **not starting from zero**:

- Master's thesis in computational biology.
- Published research in metabolic modeling.
- Spring Boot, Spring Security, JWT, microservices.
- Interest in building an AI-driven healthcare company.
- Ability to invest significant time and money into learning.
- Existing software engineering experience.

This changes the strategy completely.

You should **not** prepare like a fresh graduate trying to crack DSA interviews.

Your goal should be:

> Become an AI-enabled Senior Software Engineer / AI Full Stack Engineer who can clear MAANG-level interviews today and remain highly employable through 2031.

The market is increasingly converging software engineering, data engineering, and AI engineering into one role. Companies are looking for engineers who can build complete AI-powered products rather than specialists who only write CRUD APIs. Business Insider +1

Target Role (2026–2031)

Focus on:

Primary

AI-Enabled Full Stack Engineer

Secondary

- Senior Backend Engineer
- AI Engineer
- Forward Deployed Engineer
- Staff Software Engineer

Demand for engineers who can combine software engineering with AI systems is rising across the US, India, and Australia. AI-enabled engineering and forward-deployed engineering roles are among the fastest-growing categories. The Times of I… +1

What MAANG Looks For Today

Layer 1 — Core CS Foundations

These are still mandatory.

DSA

Master:

- Arrays
- Strings
- Linked Lists
- Trees
- BST
- Graphs
- Heaps
- Tries
- Dynamic Programming
- Greedy
- Backtracking

Target:

- 300–400 LeetCode problems
- Medium difficulty should become routine

Time:

4–6 months.

Computer Science Fundamentals

Master:

- Operating Systems
- Networking
- Databases
- Distributed Systems
- Concurrency

Books:

- Operating Systems: Three Easy Pieces
- Designing Data Intensive Applications
- Computer Networking: Top Down Approach

Layer 2 — System Design

This is where experienced engineers get selected.

Master:

- Scalability
- Caching
- Load balancing
- Sharding
- Event-driven systems
- Kafka
- CAP theorem
- Consistency models
- Distributed transactions

Build designs for:

- WhatsApp
- Uber
- YouTube
- Instagram
- Netflix

Book:

Designing Data-Intensive Applications

Layer 3 — Cloud Engineering

Mandatory.

Master:

AWS

- EC2
- ECS

- EKS
- Lambda
- RDS
- DynamoDB
- S3
- IAM

Kubernetes

- Pods
- Deployments
- Services
- Ingress
- Helm
- Observability

DevOps

- Docker
- GitHub Actions
- Terraform
- CI/CD

Layer 4 — AI Engineering

This is the most important new area.

Many engineers are learning prompting.

Few engineers can build production AI systems.

Become the second type.

AI Foundations

Learn deeply:

- Linear Algebra
- Probability
- Statistics
- Optimization

Because of your research background, this should be easier than for most software engineers.

Machine Learning

Master:

- Regression
- Classification
- Clustering
- Feature Engineering
- Evaluation Metrics

Libraries:

- NumPy
- Pandas
- Scikit-Learn

Deep Learning

Master:

- Neural Networks
- CNNs
- RNNs
- Transformers
- Attention

Frameworks:

- PyTorch
- TensorFlow

PyTorch should be primary.

Layer 5 — LLM Engineering

This is currently the hottest skill area.

Learn:

- Prompt Engineering
- RAG
- Vector Databases
- Embeddings
- Agents

- Tool Calling
- MCP
- Evaluation
- Guardrails
- Fine Tuning

Tools:

- LangGraph
- LangChain
- OpenAI SDK
- Anthropic SDK
- LlamaIndex

Companies increasingly expect engineers to integrate AI into products. Reuters +1

Layer 6 — Agent Engineering

This is likely the highest-growth area for the next five years.

Build:

Single Agent

- Research assistant

Multi-Agent

- Medical diagnosis assistant
- Healthcare monitoring workflow
- Autonomous coding assistant

Learn:

- Agent orchestration
- Planning
- Memory
- Tool usage
- Evaluation

Microsoft and others are heavily investing in autonomous AI agents. Reut…

Layer 7 — AI-Native Software Engineering

This is where the industry is heading.

Use daily:

- Cursor
- Claude Code
- GitHub Copilot
- OpenAI Codex

Learn:

- AI-assisted architecture
- AI-assisted testing
- AI-assisted debugging
- Agentic development workflows

Modern engineering increasingly involves supervising AI systems rather than writing every line manually. IT ... +1

Layer 8 — Healthcare AI Specialization

This is your unfair advantage.

Most engineers will learn AI.

Very few will understand:

- Healthcare workflows
- Diagnostics
- Biomedical data
- Clinical decision systems
- Medical regulations

Build expertise in:

Medical AI

- Medical Imaging
- Clinical NLP
- Wearable Data Analysis
- Predictive Healthcare

Standards

- FHIR
- HL7

Compliance

- HIPAA
- GDPR
- Indian DPDP Act

This aligns directly with your startup vision.

Projects To Build

These matter more than certificates.

Project 1

AI-Powered Interview Assistant

Skills:

- RAG
- Agents
- Evaluation

Project 2

AI Coding Assistant

Skills:

- MCP
- Tool Calling
- Multi-agent architecture

Project 3

Healthcare Copilot

Skills:

- Medical RAG
- Clinical reasoning
- Report generation

Project 4

Healthcare Monitoring Platform

Skills:

- Spring Boot
- Kafka
- Kubernetes
- AI inference

This directly supports your startup idea.

Interview Preparation Plan

Months 1–4

- DSA
- LeetCode
- OS
- DBMS
- Networking

Daily:

- 2–3 hours

Months 5–8

- System Design
- Distributed Systems
- AWS
- Kubernetes

Build:

- Production microservices

Months 9–12

- ML
- Deep Learning
- PyTorch

Build:

- Real models

Months 13–18

- LLM Engineering
- Agent Systems
- AI Products

Build:

- Healthcare AI projects

5-Year Career Outlook

The engineers most likely to command top compensation by 2031 will combine:

1. Strong software engineering
2. Distributed systems expertise
3. AI engineering
4. Product thinking
5. Domain expertise

Pure frontend developers, pure CRUD backend developers, and pure prompt engineers are likely to face the most pressure from AI automation. Companies increasingly value engineers who can own complete AI-enabled systems. Business Insi... +2

Recommended Priority Order For You

1. DSA + Interview Preparation
2. System Design
3. Kubernetes + Cloud
4. Python for AI
5. ML Fundamentals
6. Deep Learning
7. LLM Engineering
8. Agent Engineering
9. Healthcare AI
10. Leadership and Product Thinking

Given your background, I would target becoming a **Senior AI-Enabled Full Stack Engineer with Healthcare AI expertise** rather than a pure ML researcher. That path keeps you competitive for MAANG-style companies, AI-first startups, and also directly accelerates your long-term healthcare startup vision.